

Task 1 – CIA Examples in Cloud Applications

Principle Application Example

Confidentiality Google Drive Only authorized users can access a private document

Confidentiality Instagram Private account content is restricted to approved followers

Confidentiality Spotify Account information is protected from unauthorized access

Integrity Google Drive A document should not be silently modified by an unauthorized person

Integrity Spotify Playlist and account information should remain accurate

Integrity Instagram User profile information should not be modified without authorization

Availability Google Drive Users should be able to access stored files when needed

Availability Spotify Users should be able to stream music when the service is operational

Availability Instagram Users should be able to access posts and messages when the service is available

Task 2 – Five Common Cloud Security Threats

Threat Description Mitigation

Data Breach Unauthorized access to sensitive information Encryption, access control and monitoring

DDoS Attack Large volumes of traffic overwhelm a service DDoS protection, rate limiting and traffic filtering

Insecure APIs Poorly protected APIs may expose cloud resources Authentication, authorization and API security testing

Account Hijacking Attackers obtain user credentials MFA, strong passwords and credential monitoring

Data Loss Data may be accidentally deleted or become unavailable Backups, versioning and disaster recovery

Task 3 – AWS Management Console

Use the AWS Management Console:

AWS Management Console

After signing in:

Open the AWS Management Console.
Check the account information.
Select a region from the region selector.
Take a screenshot of the dashboard.
Insert the screenshot into your assignment.
Label These Sections

Your screenshot can be labelled with:

1. AWS Services
 2. Search Bar
 3. Region Selector
 4. Account Information
 5. Recently Visited Services
 6. AWS Service Categories
- Screenshot Caption

Figure 1: AWS Management Console Dashboard showing the selected AWS Region and account information.

Important: Before submitting, hide sensitive information such as account IDs, access keys, billing information, or other credentials if they are visible.

Task 4 – Create a User With Basic Read Permissions

AWS IAM is used to manage identities and permissions.

Open:

[AWS IAM Documentation](#)

Steps

Sign in to AWS.

Search for IAM.

Open IAM.

Select Users.

Select Create user.

Enter a username, for example:

security-read-user

Configure the required authentication method.

On permissions, select a read-only policy appropriate for the lab.

Do not assign administrator privileges.

Review the configuration.

Create the user.

Example Permission

For a simple demonstration, a read-only policy such as:

ReadOnlyAccess

can be used if appropriate for the assignment environment.

Security Principle

This demonstrates the Principle of Least Privilege: a user should receive only the permissions required to perform their job.

Task 5 – Real-World Cloud Security Incident

AWS S3 Credential Abuse and Ransomware Activity

AWS reported in January 2025 that its incident-response teams detected an increase in unusual encryption activity involving Amazon S3 buckets. The activity involved attackers using valid customer credentials to perform large numbers of S3 CopyObject operations with customer-provided encryption keys, effectively re-encrypting objects. AWS stated that the

activity did not exploit a vulnerability in the AWS service itself; instead, unauthorized actors had obtained valid customer credentials. AWS recommended short-term credentials, recovery procedures, monitoring for unexpected access patterns, and blocking SSE-C when it is not required.

Threat

Compromised Cloud Credentials

How It Was Exploited

Exposed/compromised credentials



Unauthorized AWS access



S3 API operations



Objects re-encrypted



Potential data availability/ransom impact

Prevention

The incident demonstrates the importance of:

Short-lived credentials

IAM least privilege

MFA

Credential rotation

CloudTrail monitoring

GuardDuty/security monitoring

Backup and recovery procedures

Monitoring unusual S3 activity

AWS also reported that exposed long-term credentials continue to be a major entry point observed in its incident-response work.

Practical Commands – Quick Reference

SSL/TLS

```
sudo apt update
sudo apt install apache2 openssl -y
```

```
sudo mkdir -p /etc/apache2/ssl
```

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/apache2/ssl/localhost.key \
-out /etc/apache2/ssl/localhost.crt
```

```
sudo a2enmod ssl
sudo a2ensite default-ssl
sudo systemctl restart apache2
```

Open:

```
https://localhost
```

SHA-256

```
printf '%s' 'ZomatoOrder2024' | openssl dgst -sha256 > zomato_sha256.txt
```

MD5

```
openssl dgst -md5 my_playlist.txt
```

SHA-256

```
openssl dgst -sha256 my_playlist.txt
```

Digital Signature

```
openssl genpkey -algorithm RSA -out private_key.pem
```

```
openssl pkey -in private_key.pem -pubout -out public_key.pem
```

```
openssl dgst -sha256 \
```

```
-sign private_key.pem \
```

```
-out flipkart_order.sig \
```

```
flipkart_order.txt
```

```
openssl dgst -sha256 \
```

```
-verify public_key.pem \
```

```
-signature flipkart_order.sig \
```

flipkart_order.txt

Expected:

Verified OK